MIDTERM 2 REVIEW SHEET
MATH 411: HONORS COMPLEX VARIABLES, FALL 2026

Date:	Friday, November 6, 2026, in class (CAB 369, 3:00–3:50 PM)
Length:	50 minutes
Format:	Closed book. No notes, no textbook, no calculators or devices.
Coverage:	Lectures 13–23; Lang §§3.3–3.7, 4.1, 5.2–5.3, 6.1–6.2

Note: material from Lecture 24 (argument principle, Rouché's theorem, open mapping theorem) will *not* be tested.

TOPICS

1. Primitives, Cauchy's theorem on a disc, and contour shifts (Lectures 13–15; Lang §3.3).

- A *primitive* for continuous $f : U \to \mathbb{C}$ is a holomorphic $g' = f$ (analytic functions have local primitives by termwise integration). Then $\int_\gamma f = g(\gamma(b)) - g(\gamma(a))$; on connected U, $\int_\gamma f = 0$ for closed γ iff f has a primitive ($1/z$ on $\mathbb{C} \setminus \{0\}$ has none, as $\int_{|z|=1} dz/z = 2\pi i$).
- **Primitives and Cauchy's theorem on a disc:** if f is continuous on a disc D about z_0 and $\int_{\partial R} f = 0$ for rectangles $R \subseteq D$, then $g(z_1) = \int_{z_0}^{z_1} f$ along a horizontal-then-vertical path is a primitive for f (know the proof via $\int_{\partial R} f = 0$, $f(z) = f(z_1) + \psi(z)$, and the ML-estimate). By Cauchy–Goursat, every holomorphic f on a disc D has a primitive, so $\int_\gamma f = 0$ for closed γ in D.
- Computing real integrals via Cauchy's theorem and the ML-estimate: shifting rectangular contours (the Fourier transform $\int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x\xi}\, dx = e^{-\pi\xi^2}$), sector contours (e.g. $\int_0^\infty \cos(t^2)\, dt$), and indented semicircles ($\int_0^\infty \frac{1-\cos x}{x^2}\, dx = \frac{\pi}{2}$ by splitting $\frac{1-e^{iz}}{z^2} = -\frac{i}{z} + E(z)$ with E entire).

2. Continuous curves, homotopy, and Cauchy's formula (Lectures 15–18; Lang §§3.4–3.7).

- Compactness in $\mathbb{C}$: finite subcover lemma (bisecting squares), uniform neighbourhood lemma ($D(z,r) \subseteq U$ for $z \in K$), and the **chain-of-discs lemma**: any continuous $\gamma : [a,b] \to U$ admits a *partition* $(a = a_0 < \cdots < a_n = b; D_0, \ldots, D_{n-1})$ with $\gamma([a_j, a_{j+1}]) \subseteq D_j \subseteq U$.
- For holomorphic f and continuous γ, set $\int_\gamma f = \sum_{j=0}^{n-1} [g_j(\gamma(a_{j+1})) - g_j(\gamma(a_j))]$ with primitives g_j on D_j. Know why this is well-defined ($g_j - h_\ell$ is constant on overlaps $D_j \cap \widetilde{D}_\ell$ of a common refinement) and agrees with the path integral for piecewise C^1 γ.
- **Homotopy with fixed endpoints** $H : [a,b] \times [0,1] \to U$; straight-line homotopy in convex sets; **null-homotopic** closed curves and **simply connected** domains (nonempty, connected, every loop null-homotopic). Convex and star-shaped open sets are simply connected; $\mathbb{C} \setminus \{0\}$ is not.
- **Homotopy form of Cauchy's theorem:** if $\gamma \sim \eta$ with fixed endpoints in U and f is holomorphic on U, then $\int_\gamma f = \int_\eta f$. Know the proof: *close* curves (sharing a partition) have equal integrals as $g_{j-1} - g_j = C_j$ on $D_{j-1} \cap D_j$ telescopes, and a fine grid on $[a,b] \times [0,1]$ makes consecutive slices $H(\cdot, s_k)$ close. Hence $\int_\gamma f = 0$ if γ is null-homotopic, and on simply connected U every f has a **global primitive** $g(z) = \int_{z_0}^{z} f(\zeta)\, d\zeta$.

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- **Local Cauchy integral formula:** for $\overline{D(c,R)} \subseteq U$, f holomorphic on U, $z_0 \in D(c,R)$, and $\gamma = \partial D(c,R)$, $f(z_0) = \frac{1}{2\pi i}\int_\gamma \frac{f(z)}{z-z_0}\,dz$ (know the proof comparing γ and C_r about z_0 via $\gamma * \alpha \sim \alpha * C_r$ in $U \setminus \{z_0\}$, applied to $\frac{f(z)-f(z_0)}{z-z_0}$ and $\frac{1}{z-z_0}$).

3. Complex logarithms and branches (Lecture 19; Lang §3.6).

- A *branch of* $\log z$ on $U \subseteq \mathbb{C}\setminus\{0\}$ is a holomorphic L with $e^{L(z)} = z$ ($L'(z) = 1/z$); no continuous $\log z$ or $z^{1/n}$ ($n \geq 2$) exists on $\mathbb{C}\setminus\{0\}$. On simply connected $U \subseteq \mathbb{C}\setminus\{0\}$ ($e^{w_0} = z_0 \in U$), $L(z) = w_0 + \int_{z_0}^z d\zeta/\zeta$ is a branch of $\log z$. **Principal branch** $\operatorname{Log} z = \ln|z| + i\operatorname{Arg} z$ ($-\pi < \operatorname{Arg} z < \pi$) on $\mathbb{C}\setminus(-\infty,0]$; branches on $U_\theta = \mathbb{C}\setminus\{re^{i\theta} : r \geq 0\}$ and of $z^\alpha = e^{\alpha L(z)}$, $z^{1/n} = e^{L(z)/n}$.
- **Logarithms of nonvanishing functions:** on simply connected U, if $f \neq 0$ is holomorphic, $L_f(z) = w_0 + \int_{z_0}^z (f'/f)\,d\zeta$ ($e^{w_0} = f(z_0)$) gives $e^{L_f} = f$, unique up to $2\pi i\mathbb{Z}$. Know when $L_{f_1 f_2} = L_{f_1} + L_{f_2}$ ($\operatorname{Log}(zw)$ vs. $\operatorname{Log} z + \operatorname{Log} w$) and $L(z) = L(z_0) + \sum_{n\geq 1} \frac{(-1)^{n-1}}{nz_0^n}(z-z_0)^n$ ($|z-z_0| < |z_0|$).

4. Cauchy estimates, Liouville's theorem, and Morera's theorem (Lectures 20–21; Lang §3.7).

- **Holomorphic ⇒ analytic and Cauchy estimates:** if f is holomorphic on U and $\overline{D(z_0,R)} \subseteq U$, then $f(z) = \sum_{n=0}^\infty a_n(z-z_0)^n$ on $D(z_0,R)$ (uniformly on $\overline{D(z_0,s)}$, $s < R$), with $a_n = \frac{f^{(n)}(z_0)}{n!} = \frac{1}{2\pi i}\int_{\gamma_R} \frac{f(\zeta)}{(\zeta-z_0)^{n+1}}\,d\zeta$, $|a_n| \leq \|f\|_{\gamma_R}/R^n$, and $|f^{(n)}(z_0)| \leq n!\,\|f\|_{\gamma_R}/R^n$. Know the proof: expand $\frac{1}{\zeta - z} = \sum_{n=0}^\infty \frac{(z-z_0)^n}{(\zeta-z_0)^{n+1}}$ for $|\frac{z-z_0}{\zeta-z_0}| \leq \frac{s}{R} < 1$ and integrate termwise on γ_R.
- **Liouville's theorem and the fundamental theorem of algebra:** every bounded entire function is constant ($|a_n| \leq M/R^n \to 0$ for $n \geq 1$); an entire f with $|f(z)| \leq C|z|^k$ on circles $R_j \to \infty$ is a polynomial of degree $\leq k$. Every nonconstant polynomial $P(z) = a_n z^n + \cdots + a_0$ has a root in $\mathbb{C}$ (otherwise $1/P$ is entire and bounded since $|P(z)| \geq \frac{|a_n|}{2}|z|^n$ for large $|z|$).
- **Morera's theorem:** if f is continuous on U and $\int_{\partial Q} f = 0$ for every closed rectangle $Q \subseteq U$, then f is holomorphic on U (locally $f = g'$ for a holomorphic, hence analytic, primitive g).

5. Laurent series, singularities, and residues (Lectures 21–23; Lang §§4.1, 5.2–5.3, 6.1–6.2).

- **Laurent expansion:** for f holomorphic on $A(r,R) = \{r < |z-z_0| < R\}$, $f(z) = \sum_{n\in\mathbb{Z}} a_n(z-z_0)^n = f_+ + f_-$ (uniformly on $s \leq |z-z_0| \leq S$) with unique $a_n = \frac{1}{2\pi i}\int_{|\zeta-z_0|=\rho} \frac{f(\zeta)}{(\zeta-z_0)^{n+1}}\,d\zeta$. Know the proof via Cauchy's formula on an annulus, $f(z_0) = \frac{1}{2\pi i}(\int_{C_R} - \int_{C_r})\frac{f(\zeta)}{\zeta-z_0}\,d\zeta$, and geometric series on C_R, C_r; be able to compute Laurent series on annuli via partial fractions.
- **Isolated singularities** at z_0 via $f(z) = \sum_{n\in\mathbb{Z}} a_n(z-z_0)^n$ on $0 < |z-z_0| < \rho$: *removable* ($a_n = 0$ for $n < 0 \Leftrightarrow f$ bounded near z_0), *pole of order* m ($a_{-m} \neq 0$, $a_n = 0$ for $n < -m \Leftrightarrow |f(z)| \to \infty$), *essential* ($a_n \neq 0$ for infinitely many $n < 0 \Leftrightarrow$ dense image near z_0).
- **Winding number** $w(\gamma,\alpha) = \frac{1}{2\pi i}\int_\gamma \frac{dz}{z-\alpha} \in \mathbb{Z}$ (know the proof) and **residue** $\operatorname{Res}_{z_0}(f) = a_{-1} = \frac{1}{2\pi i}\int_{|z-z_0|=r} f(z)\,dz$. **Residue theorem:** for f holomorphic on $U \setminus \{z_1,\ldots,z_n\}$ and γ closed and null-homotopic in U, $\int_\gamma f = 2\pi i \sum_{j=1}^n w(\gamma,z_j)\operatorname{Res}_{z_j}(f)$. Know the proof ($g = f - \sum f_-^{(j)}$ extends holomorphically to U, and each $f_-^{(j)}$ is integrated termwise).
- **Computing residues and real integrals:** at a simple pole, $\operatorname{Res}_{z_0}(fg) = g(z_0)\operatorname{Res}_{z_0}(f)$, and $\operatorname{Res}_{z_0}(g/f) = g(z_0)/f'(z_0)$ if $f(z_0) = 0 \neq f'(z_0)$; for an order-m pole, $(m-1)!\operatorname{Res}_{z_0}(f) = \lim_{z\to z_0}\big((z-z_0)^m f(z)\big)^{(m-1)}$. Evaluate $\int_0^{2\pi} Q(\cos\theta,\sin\theta)\,d\theta$ via $z = e^{i\theta}$, $\cos\theta = \frac{z+z^{-1}}{2}$, $\sin\theta = \frac{z-z^{-1}}{2i}$, $d\theta = \frac{dz}{iz}$, and $\int_{-\infty}^\infty f(x)\,dx = 2\pi i\sum_{\operatorname{Im} z_j > 0}\operatorname{Res}_{z_j}(f)$ when $|f(z)| \leq B/|z|^2$ in $\operatorname{Im} z \geq 0$.